CHAPTER 13 . THERMAL PROPERTIES of MATERIALS

13.1 Specific Heat Capacity

1. Def : The specific heat capacity, c, of a material is defined as the heat energy required to increase the temperature of 1 kg of that material by 1 K.

2. Unit – $Jkg^{-1}K^{-1}$

3. To raise the temperature of a material of a mass, m by $\Delta\theta$, the heat energy required is :

   $Q = mc\theta$ where : Q – Heat energy absorbed or released
   m – Mass of material
   c - Specific heat capacity of material
   θ - Temperature change of material

4. From the equation, if a material has high specific heat capacity, it required more energy to raise the same temperature for a material with low specific heat capacity.

5. Materials like Aluminium and Copper have low specific heat capacity, therefore these materials are easily heated and cooled.

6. Heat capacity, C, of a material is defined as the heat energy required to increase the temperature of the material by 1 K.

7. Unit – $J K^{-1}$

8. If the material has a mass, m, then its heat capacity, C,
   C = mass x specific heat capacity

Ex 1.
An Aluminium electric kettle has a mass of 1.5 kg when empty. The specific heat capacity of Aluminium is 900 $Jkg^{-1}K^{-1}$. Determine the heat capacity of the empty kettle.

Specific heat capacity of water is 4200 $Jkg^{-1}K^{-1}$. Determine the heat capacity for 2 kg of water.

If the Aluminium electric kettle is filled with 2 kg of water and the temperature of the water and kettle is raised from 20^{o} C to 100^{o} C, what is the amount of energy required? Comment your answer.

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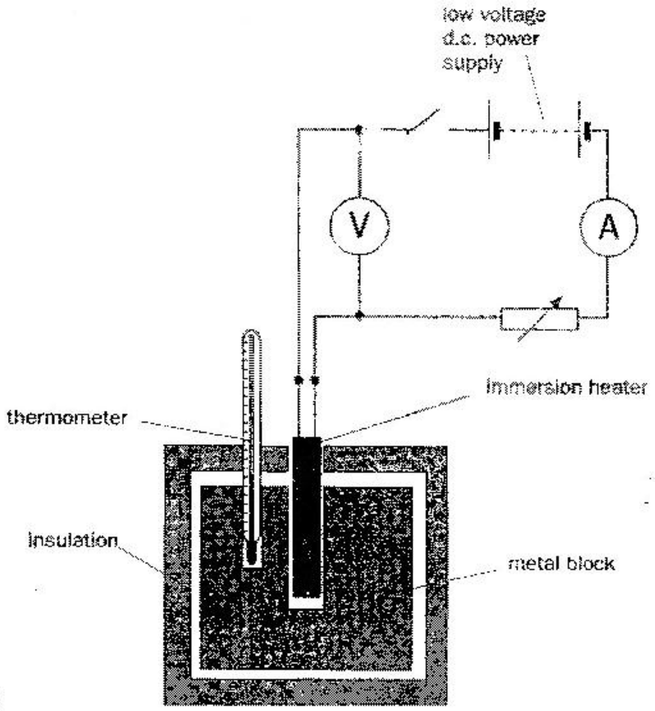
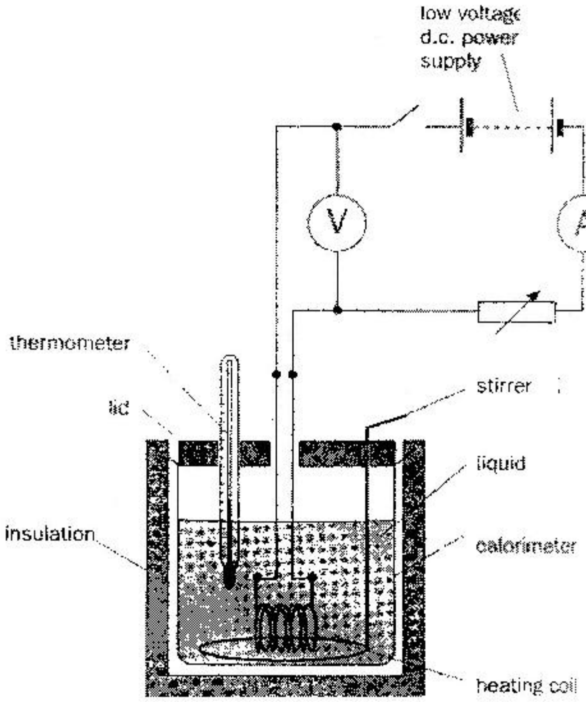
13.2 THE EXPERIMENT MEASUREMENT of SPECIFIC HEAT CAPACITY

1. Specific Heat capacity of a solid.
2. Specific Heat capacity of a liquid.





Ex 2.
A steady stream of air is drawn through a tube which contains a 500 W heater. Given that the density of air is 1.2 kgm^{-3} and its specific heat capacity is 1000 $Jkg^{-1}K^{-1}$. Calculate the maximum temperature rise of the air for a flow rate of 0.4 m^3s^{-1}.

Ex 3

A block of copper of mass 0.5 kg at an initial temperature of 77°C is placed in 0.4 kg of water at 30°C. What is the final temperature when thermal equilibrium is attained? Comment your answer. (c_w = 4200 $Jkg^{-1}K^{-1}$ c_c = 400 $Jkg^{-1}K^{-1}$)

13.3 LATENT HEAT

1. When a solid is heated, its temperature rises due to the increase of the **internal energy** of the solid.
   Recall internal energy:
2. The increase internal energy is in the form of increased *kinetic energy* of the molecules/ atom in the solid.
3. Continued heating increases the temperature of the solid to its melting point.
4. During the solid liquefy, at the melting point, the changes takes place without change of temperature.
5. Once it is completely liquefied, the temperature of the substances continues to raises again if more energy is supplied.
6. To reverse the process (from liquid to solid), energy must be taken out from the substances.
7. Energy transferred to or from a material as a result of change of state is called ***latent heat***.
8. Latent heat must be supplied for changes from :
   i) solid to liquid (melting)
   ii) liquid to vapour (boiling)
   iii) solid to vapour (sablimation)
9. The specific latent heat of fusion ( ) or vaporization ( ) of a material is defined as the energy required to change the state of 1 kg of material from solid to liquid (or liquid to vapour) without change of temperature.
10. The unit J kg^{-1}
11. To change the state of mass, *m* of a material, the energy required is given by :

$$Q = ml$$

where:
Q - energy required
m - mass of material
l - specific latent heat of fusion or vaporization of material

Ex 4

0.5 kg of water at 20° C in a plastic beaker is placed in freezer which converts it to ice in 5 minutes 30 seconds. Calculate the rate of transfer of heat from the water?
Sketch a graph Temperature – time for the situation given above.
($c_w = 4200\ Jkg^{-1}K^{-1}$ $c_i = 2000\ Jkg^{-1}K^{-1}$ and $l = 3.36 \times 10^5\ Jkg^{-1}$)

Ex 5

The temperature of a hot liquid in a container of negligible thermal capacity falls at a rate of 2° C perminute just before if begins to solidify. The temperature then remains steady for 20 minutes by which the liquid has all solidified. Determine the ratio :

$$\frac{\text{Specific heat capacity of liquid}}{\text{Specific latent heat of fusion}}$$

13.4 INTERNAL ENERGY, U

1. Internal energy of a material is the sum of kinetic energy and potential energy of all molecules in the body.

2. As the speed of the particles increases due to heat absorbed (energy gain by the material), the molecules/atoms vibrate/move at higher speed. This will increase the temperature of the body.
Recall motion of molecules/atoms in solid, liquid and gas:

3. Potential energy of a material is depends on the separation between the molecules/atoms.
Recall separation of molecules/atoms in solid, liquid and gas:
The greater the separation, the greater the Ep.

4. In the case of melting and boiling;

- E_k does not change -
- E_p changes -

* the energy supplied is used to increase the separation between the molecules/atoms (breaking up the bonds between them)

13.5 THE FIRST LAW of THERMODYNAMICS

1. The First Law of Thermodynamics is based on the law of conservation of energy.

2. Internal energy of a material is $\Delta U = \Delta Q + \Delta W$

3. When a quantity of heat, ΔQ is supplied to a fix mass of gas, its temperature may increase cause an increase in kinetic energy.

4. If the gas expands, then there is an external work done by the gas, ΔW. (recall, $\Delta W = P\Delta V$). This will cause the PE to change.

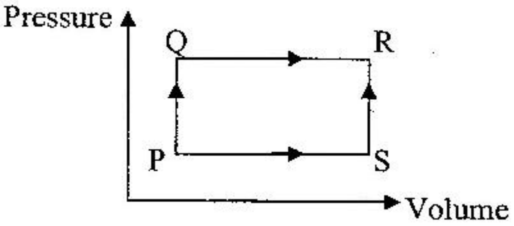
5. The combine effect of both will cause an increase/decrease in internal energy.

6. By the law of conservation of energy: $\Delta W = P(\Delta V)$

$$\Delta U = \Delta Q + \Delta W$$

Def: Increase in the internal energy of a system is equal to the heat supplied to the system (absorbed by the system) and work done on the system.

Ex 6.



$P \to S$: Work done by the gas (ΔW is -ve)
$S \to R$: No work done
$P \to Q$: No work done
$Q \to R$: Work done by the gas

When the gas is taken from state P to R by the stages PQ and QR, 8 J of heat are absorbed by it and 3 J of work are done by it. When the resultant change is achieved by stages PS and SR 1 J of work is done by the gas. Determine if heat is released or absorbed and the amount?

$PQ \to QR$

$\Delta U = 8-3$
$= 5$

$5 = \Delta Q - 1$

$\Delta Q = +6$ J

Heat is absorbed.